Machine Learning-Based RUL Estimation and Maintenance Alerting for Aircraft Turbofan Engines

Full report: report.docx (figures, tables, scorecard). This PDF is a text summary; all numbers trace to results/*.csv.

Abstract

See report.docx section Abstract.

1 Introduction

See report.docx section 1 Introduction.

6 Headline results

Capped test RMSE by model x subset (rows=models):
cols: FD001 FD002 FD003 FD004
cnn: 14.6 16.5 17.7 18.5
hgb: 12.4 12.9 13.2 13.2
hgb_tuned: 12.2 12.8 12.9 13.0
lstm: 13.8 14.9 13.2 14.8
mean: 41.9 44.9 43.7 45.6
mlp: 13.5 14.6 14.7 13.4
rf: 12.3 13.2 14.7 13.7
ridge: 15.7 16.3 15.9 18.4

7 Scorecard

C1: 12.35 / 13.46 PASS. FD001 (hgb-direct, n=13): 0.769 FAIL (narrow); FD002 (n=53): 0.906 PASS; FD003 (n=13): 1.000 PASS; FD004 (n=40): 0.900 PASS. C3 fails-within-30 recall >=0.95: 1.00/1.00/1.00/1.00 PASS; C4 median warning lead >10 cycles: 51/53/50/50 PASS.

References

12 verified references (Saxena et al. PHM08 + 11). See DOCX.